

Institutional report - Cardiac general

The obesity paradox in elderly obese patients undergoing coronary artery bypass surgery

George Le-Bert^a, Orlando Santana^{a,*}, Andrés M. Pineda^a, Carlos Zamora^a, Gervasio A. Lamas^a, Joseph Lamelas^b^aDivision of Cardiology at Mount Sinai Medical Center, Columbia University, Miami Beach, FL, USA^bDivision of Cardiothoracic Surgery, Mount Sinai Medical Center, Miami Beach, FL, USA

Received 28 September 2010; received in revised form 7 April 2011; accepted 11 April 2011

Abstract

We sought to determine whether the protective role of the ‘obesity paradox’ was present among elderly obese patients undergoing coronary artery bypass grafting (CABG) by median sternotomy. We retrospectively analyzed 1909 consecutive patients who underwent heart surgery between January 2006 and June 2009, and identified 396 patients who were ≥ 70 years of age and had isolated CABG. Subjects were divided into three groups according to their body mass index (BMI): obese (BMI ≥ 30 kg/m²), overweight (BMI 25–29.99 kg/m²) and normal (BMI 18.5–24.99 kg/m²). Of the 396 patients, 94 were obese, 167 were overweight, and 135 had a normal BMI. The composite of in-hospital complications and hospital mortality did not differ between the groups. Re-exploration for bleeding was required in none of the obese patients, which was statistically significant ($P=0.05$) compared to the other groups; otherwise there was no statistical difference for all other complications, including total length hospital stay and length of stay in the intensive care unit. Despite being labeled as higher risk candidates preoperatively, obese elderly patients undergoing CABG did not demonstrate an increased risk of postsurgical complications. We conclude that the ‘obesity paradox’ is present in this population, and they should not be excluded from receiving the benefits of CABG. © 2011 Published by European Association for Cardio-Thoracic Surgery. All rights reserved.

Keywords: Geriatric; Obesity; Coronary artery bypass grafting surgery; Outcomes

1. Introduction

An estimated one of every five deaths in the United States is caused by coronary heart disease, and 82% of these deaths are in patients over the age of 65 [1]. Indeed, patients over the age of 65 account for $>60\%$ of all acute myocardial infarctions [2]. Recent data have demonstrated that elderly patients with extensive coronary artery disease benefit from coronary artery bypass grafting (CABG) more than they do from conservative medical therapy, with an in-hospital mortality rate similar to that of younger patients [3]. Despite this information, ‘old age’ still prevents many patients from being referred for cardiac surgery [3].

Obesity and its comorbidities are also a recognized risk factor for the development of coronary artery disease, poor cardiovascular health and shortened lifespan [4]. Currently, 33.9% of the US population is considered obese, with a body mass index (BMI) ≥ 30 kg/m² [1]. These obese patients traditionally have been labeled as higher risk candidates for CABG, and have been turned down for surgery [5].

Paradoxically, although obesity is associated with numerous coronary risk factors and comorbidities, it may play a

protective role for patients with coronary artery disease after revascularization with CABG [6, 7] or percutaneous coronary intervention [8], or even for patients on medical therapy alone [9]. Gruberg et al. described this protective association as the ‘obesity paradox’ [8].

The elderly obese, as a population, are rapidly growing, and inevitably the number of those undergoing CABG surgery will increase in the near future [10]. The lack of representation in the current medical literature involving this subset of patients warrants further analysis regarding the impact of obesity on the postoperative outcomes in this age group. Therefore, we assessed whether the protective role of the obesity paradox is present among elderly (≥ 70 years) obese (BMI ≥ 30 kg/m²) patients undergoing CABG when compared to elderly patients with an overweight or normal BMI.

2. Methods

We retrospectively analyzed a total of 1909 consecutive patients in our cardiac surgery database between 5 January 2006 and 24 June 2009 at Mount Sinai Medical Center (Miami Beach, Florida, USA). All data were entered in the Hospital’s Society of Thoracic Surgeons’ database. The hospital’s Institutional Review Board approved the research protocol and waived the requirement of informed consent.

*Corresponding author. Echocardiography Laboratory, Mount Sinai Medical Center, 4300 Alton Road, Miami Beach, FL 33140, USA. Tel.: +1-305-674-2168; fax: +1-305-674-2368.

E-mail address: osantana@msmc.com (O. Santana).

The study population included patients who underwent isolated CABG surgery by traditional median sternotomy. Any subjects that had prior cardiac surgery or required additional cardiovascular surgical repair or replacement procedures were excluded. Patients were selected on the basis of their age and BMI. All were ≥ 70 years of age. Grouping was done in accordance with the international classification of weight by the World Health Organization [6]. Patients who were of normal weight (BMI 18.5–24.99 kg/m²), overweight (BMI 25–29.99 kg/m²) or obese (BMI ≥ 30 kg/m²) were included for analysis. Underweight (BMI < 18.5 kg/m²) patients were excluded.

The data used included records and events from eight cardiothoracic surgeons during the study period. Most surgeries were performed using cardiopulmonary bypass. The methods of cardioprotection and cardioplegia varied according to operator preference. The composite primary end point for all analyses was the occurrence of any of the following: in-hospital mortality, stroke, re-exploration for bleeding, renal failure, wound infection, reintubation and prolonged ventilation. Other outcomes that were analyzed but not included in the primary end point were atrial arrhythmias, pneumonia, perioperative myocardial infarction, total intensive care unit length of stay and postoperative length of stay.

Clinical events were defined according to standard Society of Thoracic Surgeons definitions. Stroke was defined as a new focal neurological deficit that persisted > 24 hours after its onset. In-hospital mortality was defined as death during the same hospital admission regardless of cause.

2.1. Statistical analysis

The data from the continuous and categorical variables were recorded as percentages, means $\pm$ standard deviation, or medians with interquartile range of the 25th and 75th

percentiles. Comparisons were made between BMI groups with the χ^2 and Mann–Whitney tests as appropriate. Logistic regression analysis were performed to evaluate for the influence of patient characteristics on end points. A *P*-value of < 0.05 was considered statistically significant. The statistical analysis was performed using SPSS version 17.0 (SPSS Inc, Chicago, IL, USA).

3. Results

A total of 396 patients were identified who underwent isolated CABG within the index time period and were within the BMI ranges for analysis. The mean age was similar between groups. The obese group had a higher prevalence of diabetes mellitus ($P=0.002$) and atrial arrhythmias ($P=0.02$), while the overweight and normal BMI patients were more likely to have chronic obstructive pulmonary disease ($P=0.001$). Medical therapy differed between groups in that more obese subjects were on angiotensin-converting enzyme inhibitors. The other patient factors were similar between groups (Table 1).

There were no observed differences in the operative variables between groups, including the use of bilateral internal mammary artery grafts (Table 2). The normal-BMI group tended to have more emergent CABG, but the difference was not statistically significant. The composite end point and in-hospital mortality numbers failed to demonstrate any statistical difference between groups (Table 3). Obese patients were less likely to require re-exploration for bleeding ($P=0.05$). Wound infections at all surgical sites were grouped. The obese patients did not have any wound infections throughout their stay.

Multiple logistic regression analysis were performed in both forward and backward stepwise fashions and failed to demonstrate any variables that had significant impact on end points.

Table 1. Baseline patient and treatment characteristics

Variable	Normal BMI=18.5–24.99 kg/m ²	Overweight BMI 25–29.99 kg/m ²	Obese BMI ≥ 30 kg/m ²	<i>P</i> -value
Body mass index (kg/m ²)	23.1 (IQR 21.7–24.2)	27.3 (IQR 26.0–28.6)	32.4 (IQR 30.7–34.6)	0.02
Age (years)	77.6 $\pm$ 5.6	76.9 $\pm$ 4.5	76.9 $\pm$ 4.6	0.50
Male	84 (62.2%)	118 (70.7%)	60 (63.8%)	0.26
Cerebrovascular disease	21 (15.6%)	24 (14.4%)	9 (9.6%)	0.40
Hypertension	125 (92.3%)	157 (94%)	90 (95.7%)	0.62
Diabetes mellitus	45 (33.3%)	55 (32.9%)	50 (53.2%)	0.002
Hyperlipidemia	103 (76.3%)	100 (59.9%)	73 (77.7%)	0.001
Peripheral vascular disease	28 (20.7%)	23 (13.8%)	10 (10.6%)	0.09
Congestive heart failure	37 (27.4%)	37 (22.2%)	32 (34%)	0.11
Atrial arrhythmias	23 (17%)	27 (16.2%)	28 (29.8%)	0.02
COPD ^a	15 (11.1%)	33 (19.8%)	14 (14.9%)	< 0.001
Creatinine	1.0 (IQR 0.89–1.3)	1.1 (IQR 0.9–1.4)	1.1 (IQR 0.9–1.3)	0.72
Ejection fraction	50% (IQR 38–55)	50% (IQR 42.2–60)	50% (IQR 39.5–60)	0.39
Triple vessel coronary artery disease	92 (68.1%)	131 (78.4%)	63 (67%)	0.06
Medical therapy				
Aspirin	59 (43.7%)	88 (52.7%)	51 (54.3%)	0.19
Anticoagulation ^b	31 (23%)	50 (29.9%)	18 (19.1%)	0.12
Warfarin	3 (2.2%)	3 (1.8%)	1 (1.1%)	0.80
ACE inhibitor	51 (37.8%)	77 (46.1%)	51 (54.3%)	0.05
β -Blocker	96 (71.1%)	120 (71.9%)	78 (83%)	0.09
Lipid-lowering agent	87 (64.4%)	124 (74.3%)	64 (68.1%)	0.18

Values expressed as mean $\pm$ standard deviation, number (%) or median (25th–75th percentile). ^aCOPD: Normal, $n=118$; overweight, $n=167$; obese, $n=89$ (5.6% missing). ^bLow molecular weight heparin or unfractionated heparin at doses achieved for full anticoagulation. BMI, body mass index; COPD, chronic obstructive pulmonary disease; IQR, interquartile range.

Table 2. Operative details

Variable	Normal BMI 18.5–24.99 kg/m ² (n=135)	Overweight BMI 25–29.99 kg/m ² (n=167)	Obese BMI ≥30 kg/m ² (n=94)	P-value
Emergent surgery	11 (8.1%)	19 (11.4%)	4 (4.3%)	0.14
Cardiogenic shock	4 (3%)	8 (4.8%)	2 (2.1%)	0.49
Cross-clamp time (minutes)	49 (IQR 10–66)	32 (IQR 23–48)	49 (IQR 25–64)	0.06
Perfusion time (minutes)	80 (IQR 53–103)	62 (IQR 52–78)	82 (IQR 53–104)	0.05
Off cardiopulmonary bypass	7 (5.2%)	9 (5.4%)	6 (6.4%)	0.95
Intra-aortic balloon pump use	38 (28.1%)	45 (26.9%)	27 (28.7%)	0.95
Bilateral internal mammary artery grafts	4 (3%)	5 (3%)	4 (4.3%)	0.83

Values expressed as number (%) or median with interquartile range (25th–75th percentile). BMI, body mass index.

4. Discussion

Obesity has become the leading cause of preventable death in the United States, and is linked to multiple medical problems, including hypertension, diabetes, coronary artery disease, heart failure, atrial arrhythmias and decreased survival [4]. Paradoxically, recent trials have suggested that obese patients with coronary artery disease actually have better outcomes when compared to their leaner counterparts when undergoing CABG [6, 7]. Along with obesity, advanced age has also been identified as a risk factor for perioperative complications. However, recent evidence suggests that, in the absence of comorbidities, short-term and long-term survival rates in the elderly are comparable to those in younger patients, with age not being an isolated risk factor for cardiac surgery and not determining postoperative outcomes [3].

Some have suggested that the obesity paradox could be attributed to the fact that overweight and obese patients were typically on better medical therapy, treated more aggressively and of younger age [9]. In our study, we were able to eliminate the confounding variable of age since the mean age between obese and non-obese subjects was similar. The obese population did, however, receive more angiotensin-converting enzyme inhibitors than the other groups. Further analysis through a multivariable logistic regression model failed to demonstrate any significant independent influence of medical therapy on outcomes.

Obese patients undergoing CABG are reported to have an increased incidence of wound infections [5, 10], renal

failure [11], atrial arrhythmias [5], prolonged ventilation [10], and increased intensive care unit [10, 12] and hospital length of stay [5, 6, 10, 12]. In contrast to the published literature, our obese elderly patients did not have a higher incidence of any of these poor outcomes, including mortality, and paradoxically had a lower rate of re-exploration for bleeding. This finding of less bleeding is consistent with a prior study by Birkmeyer et al. involving more than 11,000 patients undergoing CABG, which reported a protective effect of obesity on postoperative bleeding [6]. Some have attributed this to possibly a lower degree of hemodilution using relatively fixed bypass circuit sizes, which may decrease bypass-generated coagulopathy in obese patients [13]. Our report extends this finding to the obese elderly patient.

Our study has limitations. Most trials that involved obesity and coronary revascularization demonstrated that the obesity paradox may not have a linear distribution. There may be a 'U-shaped' or parabolic distribution of its effects, where the peak benefits seem to be more prevalent in the obese groups and are reduced as the level of obesity increases [13]. If so, the mean BMI of 32.4 kg/m² for our obese group is well within the range in which there is a detectable benefit. Second, our mortality results were limited to in-hospital stay. A longer follow-up period may have detected additional events influencing the final results. Also, this is a retrospective single-center study.

In conclusion, our results suggest that obesity has at least a neutral effect on the in-hospital morbidity and mortality

Table 3. Complications

Variable	Normal BMI 18.5–24.99 kg/m ² (n=135)	Overweight BMI 25–29.99 kg/m ² (n=167)	Obese BMI ≥30 kg/m ² (n=94)	P-value
Postoperative length of stay (days)	10 (7–16)	8 (7–12)	10 (7–13)	0.44
Intensive care unit length of stay (hours)	90 (52–139)	73 (48–116)	75 (52–121)	0.80
Prolonged ventilation	44 (32.6%)	44 (26.3%)	26 (27.7%)	0.47
Reintubation	17 (12.6%)	12 (7.2%)	10 (10.6%)	0.28
Wound infection	3 (2.2%)	4 (2.4%)	0 (0%)	0.33
Pneumonia	10 (7.4%)	6 (3.6%)	1 (1.1%)	0.06
Stroke	6 (4.4%)	4 (2.4%)	1 (1.1%)	0.29
Acute renal failure	6 (4.4%)	11 (6.6%)	7 (7.4%)	0.60
Re-exploration for bleeding	7 (5.2%)	11 (6.9%)	0 (0%)	0.05
Atrial arrhythmias	48 (35.6%)	56 (33.5%)	28 (29.8%)	0.66
Perioperative myocardial infarction	0 (0%)	2 (1.2%)	0 (0%)	0.25
Composite end point	55 (40.7%)	54 (32.3%)	33 (35.1%)	0.31
In-hospital mortality	8 (5.9%)	7 (4.2%)	4 (4.3%)	0.75

Values expressed as number (%) or median (25th–75th percentile). BMI, body mass index.

of elderly patients undergoing CABG, and may in fact have a limited protective role. Therefore, CABG should not be withheld from patients with advanced age and obesity on the sole premise of being thought of as higher risk candidates than their lighter peers.

Acknowledgements

We gratefully acknowledge the efforts of Kathy I. Perez, Jaime Morey and Marilyn Joy for chart review and verification, data entry and database maintenance.

References

- [1] Lloyd-Jones D, Adams R, Carnethon M, De Simone G, Ferguson TB, Flegal K, Ford E, Furie K, Go A, Greenland K, Haase N, Hailpern S, Ho M, Howard V, Kissela B, Kittner S, Lackland D, Lisabeth L, Marelli A, McDermott M, Meigs J, Mozaffarian D, Nichol G, O'Donnell C, Roger V, Rosamond W, Sacco R, Sorlie P, Stafford R, Steinberger J, Thom T, Wasserthiel-Smoller S, Wong N, Wylie-Rosett J, Hong Y. A report from the American Heart Association Statistics Committee and Stroke Statistics Subcommittee. *Circulation* 2009;119:e21–e181.
- [2] Rich MW. Epidemiology, clinical features, and prognosis of acute myocardial infarction in the elderly. *Am J Geriatr Cardiol* 2006;15:7–11.
- [3] Wieldemann D, Bernhard D, Laufer G, Kocher A. The elderly patient and cardiac surgery — a mini review. *Gerontology* 2010;56:241–249.
- [4] Lavie CJ, Milani RV, Ventura HO. Obesity and cardiovascular disease risk factor, paradox, and impact of weight loss. *J Am Coll Cardiol* 2009;53:1925–1932.
- [5] Kuduvali M, Grayson AD, Oo AY, Fabri BM, Rashid A. Risk of morbidity and in-hospital mortality in obese patients undergoing coronary artery bypass surgery. *Eur J Cardiothorac Surg* 2002;22:787–793.
- [6] Birkmeyer NJ, Charlesworth DC, Hernandez F, Leavitt B, Marrin C, Morton J, Olmstead E, O'Connor G. Obesity and risk of adverse outcomes associated with coronary artery bypass surgery. *Circulation* 1998;97:1689–1694.
- [7] Gruberg L, Mercado N, Milo S, Boersma E, Disco C, van Es G, Lemos P, Tzvi M, Wijns W, Unger F, Beyar R, Serruys P. Impact of body mass index on the outcome of patients with multivessel disease randomized to either coronary artery bypass grafting or stenting in the ARTS trial: the obesity paradox II? *Am J Cardiol* 2005;95:439–444.
- [8] Gruberg L, Weissman N, Waksman R, Fuchs S, Deible R, Pinnow E, Ahmed L, Kent K, Pichard A, Suddath W, Satler L, Lindsay J. The impact of obesity on the short-term and long-term outcomes after percutaneous coronary intervention: the obesity paradox? *J Am Coll Cardiol* 2002;39:578–584.
- [9] Steinberg BA, Cannon CP, Hernandez AF, Pan W, Peterson ED, Fonarow GC. Medical therapies and invasive treatments for coronary artery disease by body mass: the 'obesity paradox' in the Get With The Guidelines database. *Am J Cardiol* 2007;100:1331–1335.
- [10] Villavicencio MA, Sundt TM III, Daly RC, Dearani J, McGregor C, Mullany C, Orszulack T, Puga F, Schaff H. Cardiac surgery in patients with body mass index of 50 or greater. *Ann Thorac Surg* 2007;83:1403–1411.
- [11] Prabhakar G, Haan CK, Peterson ED, Coombs LP, Cruzzavala JL, Murray GF. The risks of moderate and extreme obesity for coronary artery bypass grafting outcomes: a study from the society of thoracic surgeons' database. *Ann Thorac Surg* 2002;74:1125–1131.
- [12] Ghotkar SV, Grayson AD, Fabri BM, Dihmis WC, Pullan DM. Preoperative calculation of risk for prolonged intensive care unit stay following coronary artery bypass grafting. *J Cardiothorac Surg* 2006;1:14.
- [13] Sun X, Hill PC, Bafi AS, Garcia JM, Haile E, Corso PJ, Boyce SM. Is cardiac surgery safe in extremely obese patients (body mass index 50 or greater)? *Ann Thorac Surg* 2009;87:540–547.